COMPUTER ENGINEERING

QUEZON CITY

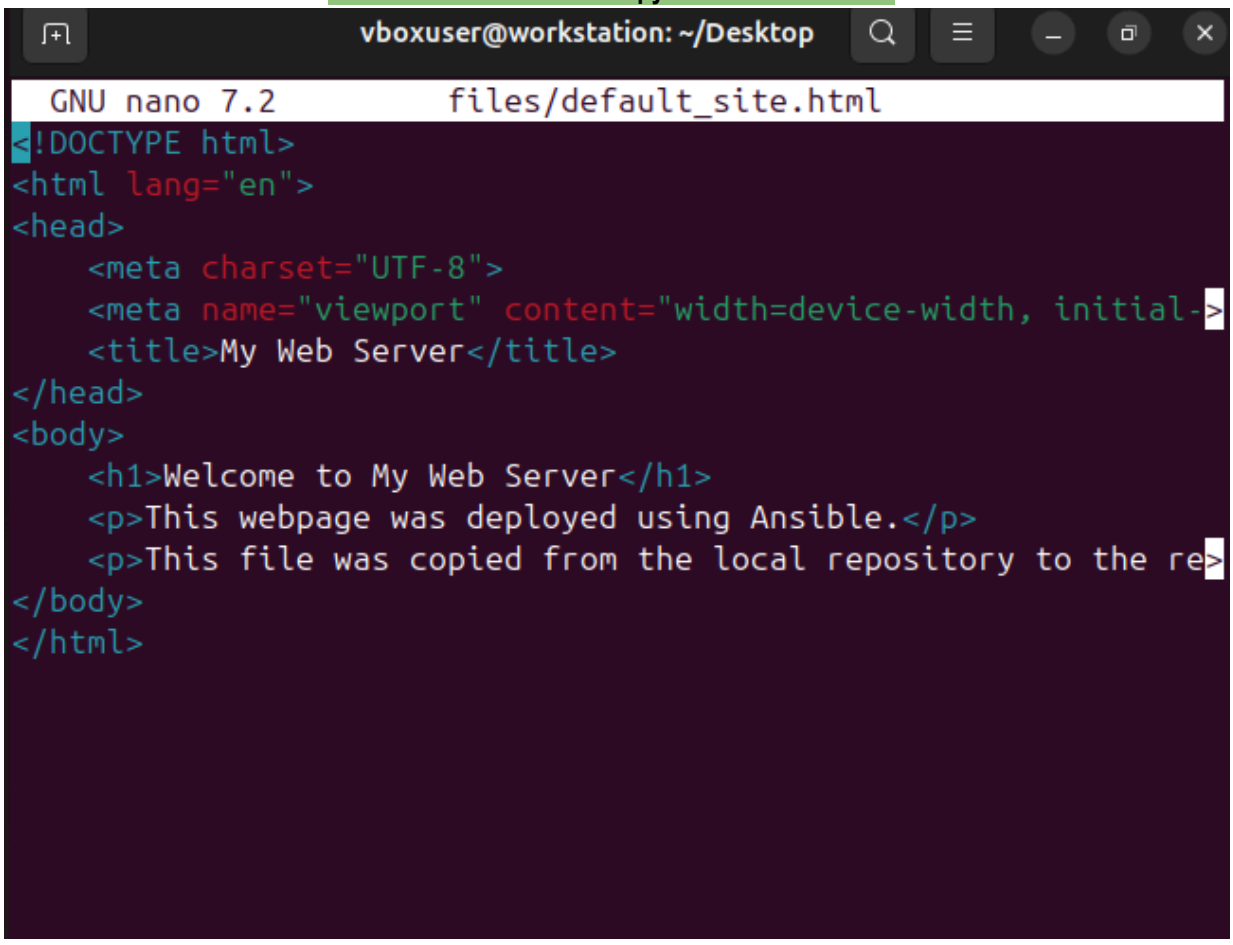
COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S3
DATE PERFORMED:	October 7, 2026	**DATE SUBMITTED:**	October 7, 2026
NAME:	Nadine A. Malabanan	**INSTRUCTOR:**	Engr. Jimlord Queado

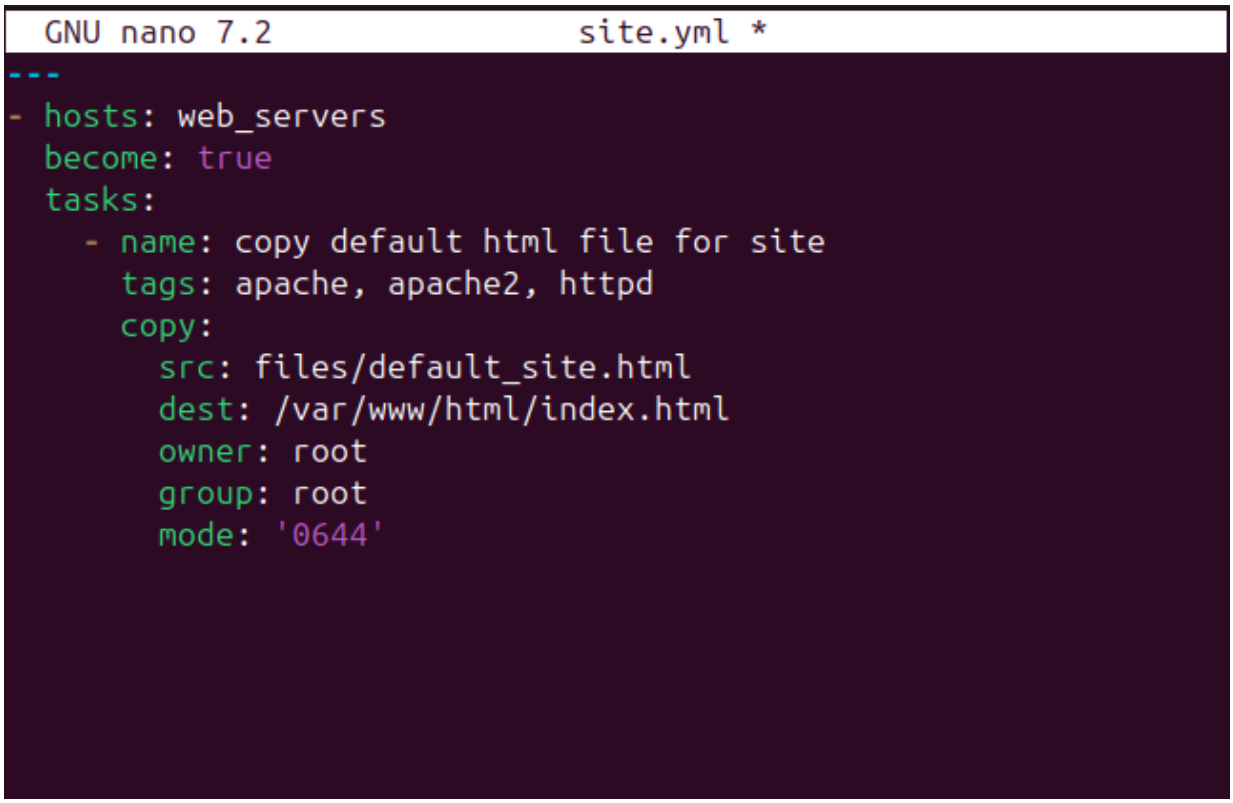
ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

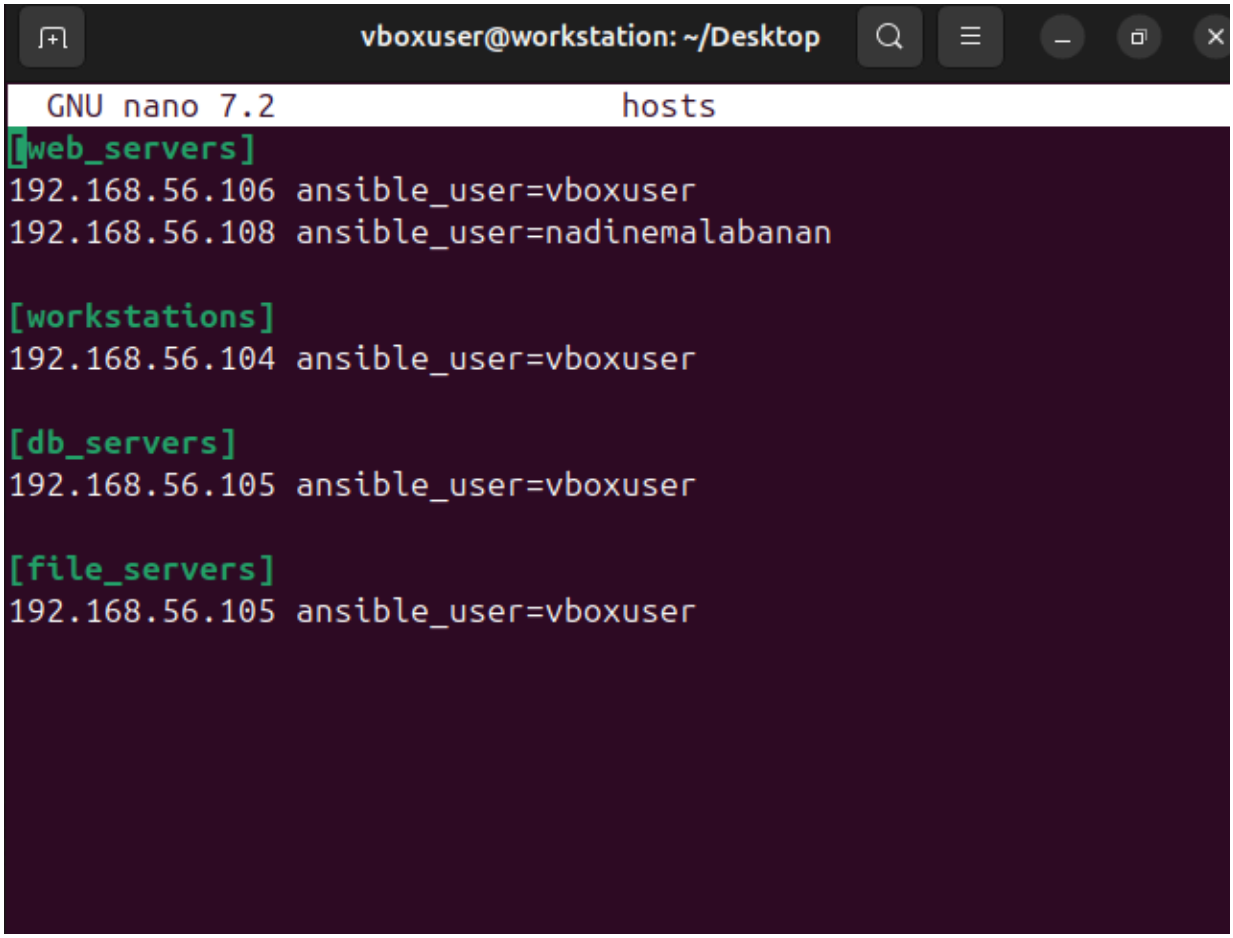
What procedures did you perform?

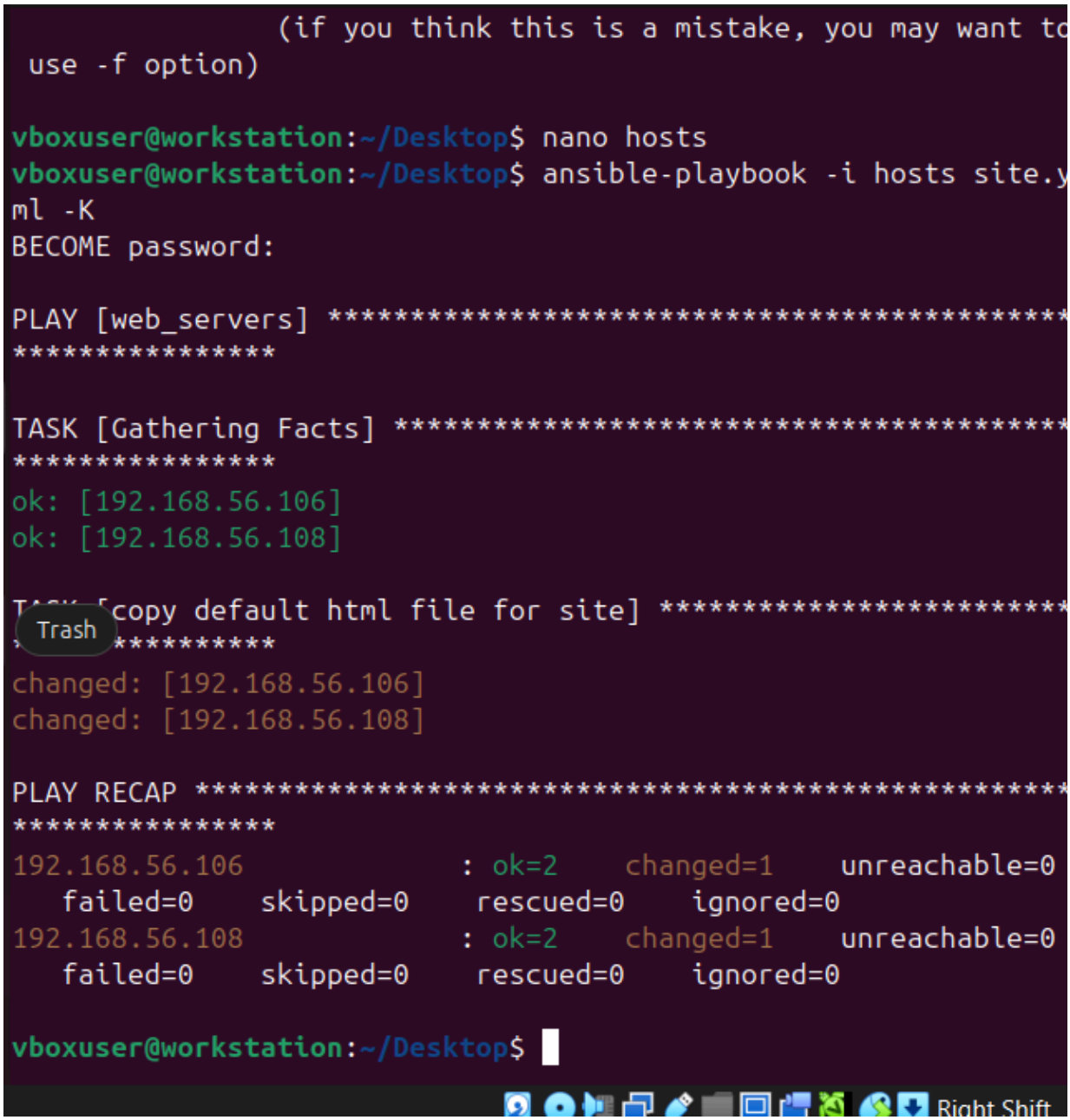
Task 1: Create a file and copy it to remote servers




GNU nano 7.2 site.yml *

- hosts: web_servers
become: true
tasks:
- name: copy default html file for site
tags: apache, apache2, httpd
copy:
src: files/default_site.html
dest: /var/www/html/index.html
owner: root
group: root
mode: '0644'


vboxuser@workstation: ~/Desktop
GNU nano 7.2 hosts
[web_servers]
192.168.56.106 ansible_user=vboxuser
192.168.56.108 ansible_user=nadinemalabanan
[workstations]
192.168.56.104 ansible_user=vboxuser
[db_servers]
192.168.56.105 ansible_user=vboxuser
[file_servers]
192.168.56.105 ansible_user=vboxuser


(if you think this is a mistake, you may want to
use -f option)
vboxuser@workstation:~/Desktop$ nano hosts
vboxuser@workstation:~/Desktop$ ansible-playbook -i hosts site.yml -K
BECOME password:
PLAY [web_servers] ****************
TASK [Gathering Facts] ****************
ok: [192.168.56.106]
ok: [192.168.56.108]
Trash
changed: [192.168.56.106]
changed: [192.168.56.108]
PLAY RECAP ****************
192.168.56.106 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
192.168.56.108 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
vboxuser@workstation:~/Desktop$
Right Shift

```
vboxuser@workstation:~/Desktop$ ssh vboxuser@192.168.56.106 "cat
 /var/www/html/index.html"
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-s
cale=1.0">
    <title>My Web Server</title>
</head>
<body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the rem
ote server.</p>
</body>
</html>
```

```
nadinemalabanan@centos-server2:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
nadinemalabanan@centos-server2:~$
```

Task 2: Download a file and extract it to a remote server


GNU nano 7.2 site.yml *

- hosts: web_servers
become: true
tasks:
- name: copy default html file for site
copy:
src: files/default_site.html
dest: /var/www/html/index.html
owner: root
group: root
mode: '0644'
- name: Install unzip package (Ubuntu)
apt:
name: unzip
state: present
update_cache: yes
when: ansible_os_family == "Debian"
- name: Install unzip package (CentOS)
yum:
name: unzip
state: present
when: ansible_os_family == "RedHat"
- name: Download Terraform binary
get_url:
url: https://releases.hashicorp.com/terraform/1.5.7/terraform_1>

```
GNU nano 7.2                        site.yml *
      mode: '0644'

  - name: Install unzip package (Ubuntu)
    apt:
      name: unzip
      state: present
      update_cache: yes
    when: ansible_os_family == "Debian"

  - name: Install unzip package (CentOS)
    yum:
      name: unzip
      state: present
    when: ansible_os_family == "RedHat"

  - name: Download Terraform binary
    get_url:
      url: https://releases.hashicorp.com/terraform/1.5.7/terraform_1>
      dest: /tmp/terraform.zip
      mode: '0644'

  - name: Unarchive Terraform binary to /usr/local/bin
    unarchive:
      src: /tmp/terraform.zip
      dest: /usr/local/bin/
      remote_src: yes
      mode: '0755'
```

```
BECOME password:

PLAY [web_servers] ************************************************
********

TASK [Gathering Facts] ********************************************
********
ok: [192.168.56.106]
ok: [192.168.56.108]

TASK [copy default html file for site] ****************************
********
ok: [192.168.56.106]
ok: [192.168.56.108]

PLAY RECAP ********************************************************
********
192.168.56.106             : ok=2    changed=0    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0
192.168.56.108             : ok=2    changed=0    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0

vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ nano site.yml
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ ansible-playbook
 -i hosts site.yml -K
BECOME password:

PLAY [web_servers] ************************************************
********

TASK [Gathering Facts] ********************************************
********
```

```
********
ok: [192.168.56.106]
ok: [192.168.56.108]

TASK [Install unzip package (Ubuntu)] ***********************************
********
skipping: [192.168.56.108]
ok: [192.168.56.106]

TASK [Install unzip package (CentOS)] ***********************************
********
skipping: [192.168.56.106]
ok: [192.168.56.108]

TASK [Download Terraform binary] ****************************************
********
changed: [192.168.56.106]
changed: [192.168.56.108]

TASK [Unarchive Terraform binary to /usr/local/bin] *********************
********
changed: [192.168.56.106]
changed: [192.168.56.108]

PLAY RECAP **************************************************************
********
192.168.56.106             : ok=5    changed=2    unreachable=0    faile
d=0    skipped=1    rescued=0    ignored=0
192.168.56.108             : ok=5    changed=2    unreachable=0    faile
d=0    skipped=1    rescued=0    ignored=0

vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$
```

Task 3: Create roles

```
  GNU nano 7.2                          site.yml *
---
- hosts: all
  become: true
  pre_tasks:

    - name: update repository index (CentOS)
      tags: always
      dnf:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations
```

Trash

```
 GNU nano 7.2                        site.yml *
      changed_when: false
      when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

GNU nano 7.2 roles/base/tasks/main.yml

```
---
- name: Install unzip package (Ubuntu)
  apt:
    name: unzip
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Install unzip package (CentOS)
  yum:
    name: unzip
    state: present
  when: ansible_os_family == "RedHat"

- name: Download Terraform binary
  get_url:
    url: https://releases.hashicorp.com/terraform/1.5.7/terraform_1.5.7
    dest: /tmp/terraform.zip
    mode: '0644'

- name: Unarchive Terraform binary to /usr/local/bin
  unarchive:
    src: /tmp/terraform.zip
    dest: /usr/local/bin/
    remote_src: yes
    mode: '0755'
```

GNU nano 7.2 roles/web_servers/tasks/main.yml *

```
---
- name: copy default html file for site
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

```
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ ansible-playbook -i hosts site.
yml -K
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.106]
ok: [192.168.56.105]
ok: [192.168.56.108]

TASK [update repository index (CentOS)] ****************************************
skipping: [192.168.56.106]
skipping: [192.168.56.105]
ok: [192.168.56.108]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.108]
ok: [192.168.56.106]
ok: [192.168.56.105]

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.108]

TASK [base : Install unzip package (Ubuntu)] ***********************************
skipping: [192.168.56.108]
ok: [192.168.56.106]
ok: [192.168.56.105]

TASK [base : Install unzip package (CentOS)] ***********************************
skipping: [192.168.56.106]
skipping: [192.168.56.105]
ok: [192.168.56.108]
```

```
skipping: [192.168.56.105]
ok: [192.168.56.108]

TASK [base : Download Terraform binary] ********************************************
ok: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.108]

TASK [base : Unarchive Terraform binary to /usr/local/bin] *************************
ok: [192.168.56.106]
ok: [192.168.56.105]
ok: [192.168.56.108]

PLAY [workstations] ****************************************************************
skipping: no hosts matched

PLAY [web_servers] *****************************************************************

TASK [Gathering Facts] *************************************************************
ok: [192.168.56.106]
ok: [192.168.56.108]

TASK [web_servers : copy default html file for site] *******************************
ok: [192.168.56.106]
ok: [192.168.56.108]

PLAY [db_servers] ******************************************************************

TASK [Gathering Facts] *************************************************************
ok: [192.168.56.105]

PLAY [file_servers] ****************************************************************

TASK [Gathering Facts] *************************************************************
ok: [192.168.56.105]

PLAY RECAP *************************************************************************
192.168.56.105             : ok=8    changed=0    unreachable=0    failed=0    skipped=
```

```
TASK [Gathering Facts] *********************************************************
ok: [192.168.56.105]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.105]

PLAY RECAP *********************************************************************
192.168.56.105             : ok=8    changed=0    unreachable=0    failed=0    skipped=
2    rescued=0    ignored=0
192.168.56.106             : ok=8    changed=0    unreachable=0    failed=0    skipped=
2    rescued=0    ignored=0
192.168.56.108             : ok=8    changed=0    unreachable=0    failed=0    skipped=
2    rescued=0    ignored=0

vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following:
1. What is the importance of creating roles?

Creating roles is important because it makes the Ansible project more organized and easier to manage. It separates different tasks and configurations, making them easier to reuse, update, and troubleshoot without affecting other parts of the project.

2. What is the importance of managing files?

Managing files is important because it helps keep server configurations consistent and secure. Ansible allows us to easily manage file contents, permissions, ownership, and settings across different servers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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